

Mohammed Saahir Essa

San Francisco Bay Area, CA | 281-736-3568 | saahiressa@gmail.com | linkedin.com/in/mohammedessa

Product Design Engineer — High-Volume Consumer Hardware & Precision Mechanisms

PROFESSIONAL EXPERIENCE

Google <i>Product Design Engineer — Next Gen Pixel</i> - Led rear camera hardware integration by engineering a unified bracket structure, enhancing thermal dissipation, structural integrity, and EMI mitigation within a more compact form factor. - Expanded battery capacity by 140mAh by championing the integration of advanced cell chemistry and an innovative pack architecture, delivering Pixel's top battery energy density by volume. - Developed a next-generation battery serviceability architecture featuring a robust interface to streamline end-user repairs. - Generated \$1.2M in lifetime savings through module standardization, upstream assembly integration, and qualification of alternate thermal materials.	San Francisco Bay Area, CA 2026 – Present
Google <i>Product Design Engineer — Pixel 10 Pro</i> - Designed lower device architecture including speaker, haptics, 3 jumper flexes, PCBA, USB-C, and microphone-barometric port; redesigned architecture enabling a 100mAh battery increase. - Elevated external aesthetics by designing a cost-effective metal mesh for the bottom acoustic ports. - Unified Pixel 10 and 10 Pro chin design to standardize components and assembly fixtures, driving \$1.5M in CapEx and piece-part cost savings.	San Francisco Bay Area, CA 2024 – 2025
Google <i>Product Design Engineer — Pixel 8</i> - Designed PCBA mechanical architecture through custom and off-the-shelf parts to achieve EMI, antenna, and thermal goals; met Google, T-Mobile, and AT&T RASS reliability standards. - Owned enclosure assembly and 6 system configurations for RF, battery, and carrier testing. - Drove \$1.4M in cost savings by standardizing internal die-cuts, validating cost-effective materials, and consolidating multi-piece stacks for FATP assembly.	San Francisco Bay Area, CA 2022 – 2023
Culture Biosciences <i>Lead Mechanical Engineer — R&D Bioreactors & Lab Hardware</i> - Lead mechanical engineer for R&D 250L pilot scale bioreactor; engineered single-use USP Class VI / FDA food-safe injection-molded plastics to ensure sterility and autoclave compatibility. - Designed ergonomic handheld device with IP43 equivalent protection for technicians to individually control 6 peristaltic pumps with 0.5 mL accuracy.	South San Francisco, CA 2021 – 2022

FEATURED ENGINEERING PROJECT

Semi-Automated Seeding Machine — Dream Harvest Farms - Developed a semi-automated device to plant seeds (2mm to 15mm) utilizing vacuum suction pressure for under \$800. - Reduced overall seeding time by 60% and decreased the required human operators from 5 to 1 for commercial hydroponics.	Capstone Design
--	-----------------

TECHNICAL SKILLS & EXPERTISE

CAD & Design: Siemens NX, Fusion 360, Onshape, Inventor, SolidWorks, GD&T, DFM/DFA, Kinematics
Manufacturing: Injection Molding, Die Casting (ZA alloy), Sheet Metal Stamping, CNC Machining, Diecut, Overmolding, PCBA
Testing & NPI: Tooling Qualification, FATP Assembly, Drop/Impact Analysis, Thermal & EMI Mitigation, Carrier Reliability